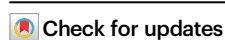


Translational regulation by oxidative desulfuration of tRNA modifications

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Modifications in the anticodon region of transfer RNA (tRNA) are essential for accurate and efficient protein synthesis. 5-Methyl-2-thiouridine derivatives (xm⁵s²U) are major modifications at the wobble position of tRNA anticodons decoding purine-ending two-codon sets. Although the thiocarbonyl group of xm⁵s²U enhances decoding efficiency, it is chemically susceptible to oxidative desulfuration, yielding 4-pyrimidinone derivatives (xm⁵h²U). Here, we identify xm⁵h²U derivatives in human cells and mouse tissues and confirm their cellular formation by spike-in experiments. Desulfurized tRNAs carrying 5-methoxycarbonylmethyl-4-pyrimidinone (mcm⁵h²U) show impaired codon recognition in a human reconstituted in vitro translation system. The mcm⁵h²U modification reduces aminoacylation of tRNAs for lysine, glutamate, and glutamine, but not arginine. Cryogenic electron microscopy reveals the structural basis of altered AAA/AAG decoding by mcm⁵h²U at the ribosomal A-site. These findings reveal a mechanism by which oxidative desulfuration of tRNA modifications dynamically regulates codon recognition and protein synthesis under oxidative stress conditions in human and mammalian cells.

Higher-order biological processes emerge from sophisticated regulation of gene expression at multiple stages at both transcriptional and post-transcriptional levels, including translational control. In protein synthesis, transfer RNA (tRNA) anticodons recognize and decode messenger RNA (mRNA) codons on the ribosome. tRNAs undergo the most extensive and diverse range of post-transcriptional modifications, particularly within the anticodon region. To date, ~150 types of RNA modifications have been identified in various RNA molecules across all domains of life, with tRNAs comprising over 80% of these modifications¹. Recent research indicates that tRNA modifications respond dynamically to cellular metabolite levels and environmental stresses, temporally and spatially influencing translational regulation significantly^{2–7}.

The anticodon of tRNA, located at positions 34, 35, and 36, recognizes a specific codon on the mRNA through base pairing at the A-site of the ribosome. In this interaction, the first and second nucleotides of the codon form Watson–Crick base pairs with the third (position 36) and second (position 35) nucleotides of the anticodon, respectively. These two Watson–Crick base pairs in the codon–anticodon helix are strictly monitored through A-minor interactions⁸ with conserved residues in the decoding center of the small ribosomal subunit (corresponding to A1492, A1493, and G530 of the bacterial 16S rRNA)⁹. These interactions trigger a large conformational rearrangement of the small subunit and induce GTP hydrolysis by the elongation factor bound to the ribosome, thereby ensuring accurate decoding during protein synthesis. By contrast, base pairing

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between the third nucleotide of the codon and the first nucleotide (position 34) of the anticodon does not necessarily follow the Watson–Crick pairing rules; instead, wobble base pairing is permitted¹⁰. This mechanism allows 61 sense codons to be efficiently decoded by a limited number of tRNAs. The first position of the anticodon often contains diverse modifications that promote or modulate codon–anticodon interactions, thereby enabling accurate and efficient decoding of the genetic code^{11–13}. Defects in tRNA

modifications frequently impair multiple stages of protein synthesis—including aminoacylation, decoding, and translocation—and can lead to pathological consequences^{2,14–16}.

2-Thiouridine (s^2U) and its derivatives (xm^5s^2U ; Fig. 1a, b) are highly conserved wobble modifications found at the first position of the anticodon in all domains of life^{17–19}. In bacteria and archaea, 5-methylaminomethyl-2-thiouridine (mnm^5s^2U) and 5-carboxymethylaminomethyl-2-thiouridine ($cmnm^5s^2U$) occur in

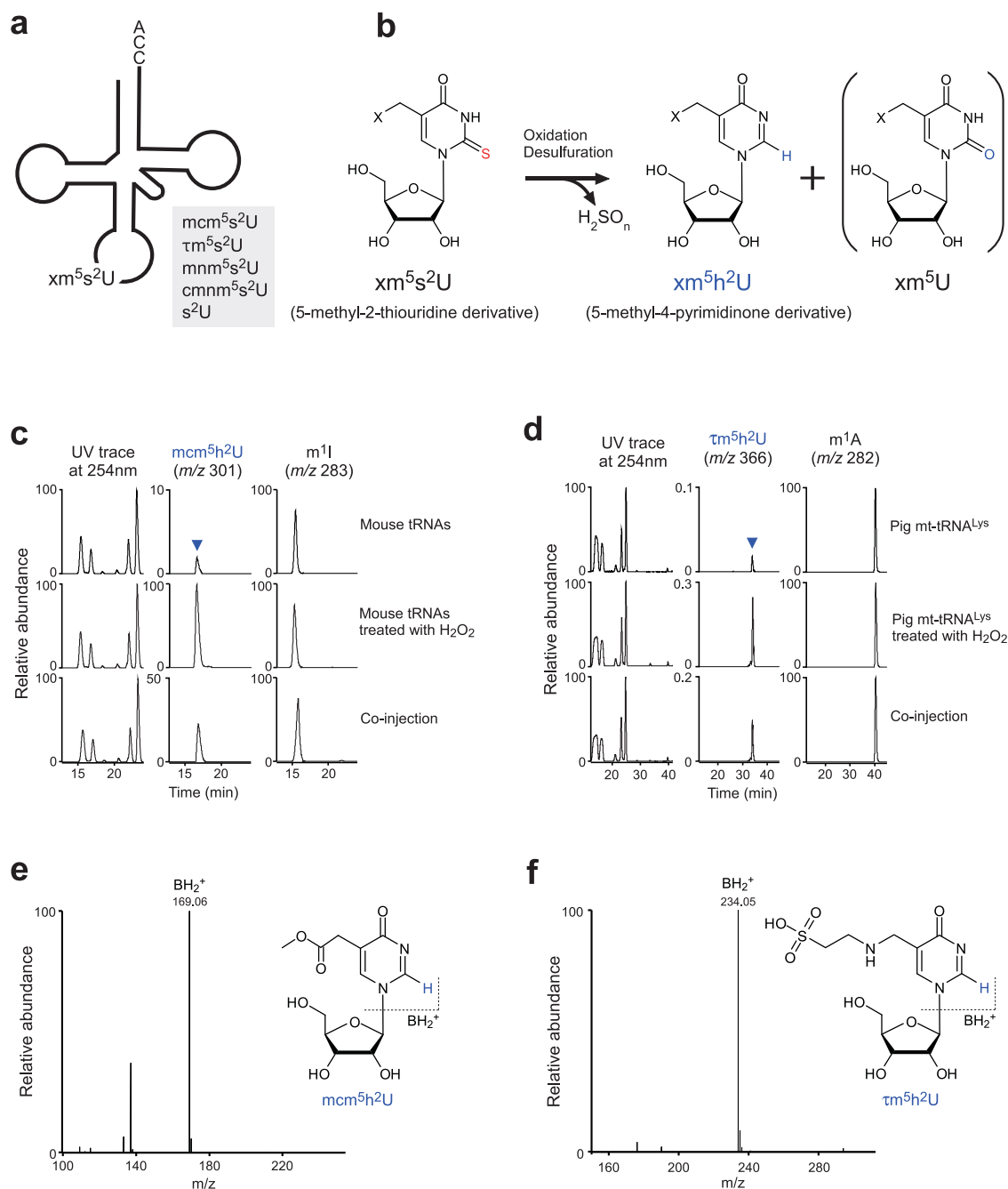


Fig. 1 | Desulfuration of 2-thiouridine derivatives found in tRNA anticodons.

a The 5-methyl-2-thiouridine derivative (xm^5s^2U) is present at the first position (34) of tRNA anticodons. mcm^5s^2U and tm^5s^2U are found in eukaryotic cytoplasmic tRNAs and animal mitochondrial tRNAs, respectively. mnm^5s^2U and $cmnm^5s^2U$ occur in bacterial cytoplasmic tRNAs. s^2U is an intermediate of xm^5s^2U . **b** Reaction scheme for oxidative desulfuration of xm^5s^2U in tRNA. 5-methyl-4-pyrimidinone derivative (xm^5h^2U) and 5-methyluridine derivatives (xm^5U) are generated by in vitro oxidation and desulfuration of xm^5s^2U . This study only focused on xm^5h^2U . **c** LC-MS nucleoside analyses of mouse total tRNA (top panels), in vitro oxidized

total tRNA (middle panels), and co-injection of both tRNAs (bottom panels) with UV traces (254 nm, left panels), mass chromatograms for mcm^5h^2U (m/z 301, center panels) and m^1I (m/z 283, right panels). The relative abundance of each peak is normalized against m^1I abundance. **d** LC-MS nucleoside analyses of pig mitochondrial tRNA^{Lys} (top panels), in vitro oxidized tRNA^{Lys} (middle panels), and co-injection of both tRNAs (bottom panels) with UV traces (254 nm, left panels), mass chromatograms for tm^5h^2U (m/z 366, center panels) and m^1A (m/z 282, right panels). The relative abundance of each peak was normalized against m^1A abundance. Collision-induced dissociation (CID) spectra of mcm^5h^2U (**e**) and tm^5h^2U (**f**).

tRNAs for Glu, Gln, and Lys. In eukaryotes, 5-methoxycarbonylmethyl-2-thiouridine (5-methylcarboxymethyl-2-thiouridine, mcm⁵s²U)²⁰ and 5-taurinomethyl-2-thiouridine (tm⁵s²U)²¹ are present in cytoplasmic and mitochondrial tRNAs for Lys, Glu and Gln, respectively^{22,23}. Mammalian cytosolic tRNA^{Arg} responsible for AGR codons also has mcm⁵s²U²⁴. Recent studies identified 2'-*O*-methyl-2-thiouridine (s²Um) and 5-cyanomethyl-2-thiouridine (cnm⁵s²U) in archaeal tRNAs^{25–27}. The s²U derivatives are also present at position 54 of tRNAs. 5-Methyl-2-thiouridine (m⁵s²U) and 1-methyl-4-thiopseudouridine (m⁵s⁴ψ) are present in thermophilic bacteria and archaea^{28–30}.

The xm⁵s²U modification performs pleiotropic roles in translation, including efficient aminoacylation of tRNAs^{31–33}, accurate and efficient decoding of purine-ending codons (NNR)^{17,34}, and rapid translocation on the ribosome³⁵. The xm⁵s²U modification favors a C3'-*endo* sugar pucker conformation due to steric repulsion between 2-thiocarbonyl and 2'-hydroxyl groups^{36,37}, so that xm⁵s²U prevents misdecoding of pyrimidine-ending codons (NNY). In addition, structural analyses revealed that mnm⁵s²U34 base pairs with the third guanosine of NNG codons at the A-site of the ribosome via an anionic U • G wobble pairing^{38,39}. In this base pairing geometry, the uracil base of xm⁵s²U34 moves toward the minor groove³⁹, rather than the major groove as occurs in canonical U • G wobble pairing¹⁰. In this geometry, 2-thio group of xm⁵s²U contributes to stacking interaction with the second base pair of the codon-anticodon helix³⁸. To make this wobble pair possible, N3 of the uracil base should be deprotonated. In fact, s²U modification promotes acidity of the N3 atom of the uracil base by lowering its pK_a value⁴⁰. This is the molecular basis by which xm⁵s²U facilitates the decoding of G-ending codons. However, according to structural study of the anticodon-stem loop bound to 30S ribosomal subunit⁴¹, the mcm⁵s²U–G base pair adopts a Watson–Crick-like geometry, and it was proposed that mcm⁵s²U may undergo a tautomeric shift of the uracil base from keto-to-enol form, allowing it to form non-canonical hydrogen bonds with G.

Biogenesis of 2-thio modification of xm⁵s²U34 has been extensively studied in both bacteria and eukaryotes^{17,18,30}. Cysteine desulfurase (bacterial IscS and eukaryotic Nfs1) initiates s²U formation by extracting a sulfur atom from cysteine as a persulfide, which is then relayed by several sulfur mediators and transferred to U34 of tRNA by thiouridylases (bacterial MnmA and the eukaryotic CTU1/CTU2 complex)^{42–46}. In bacteria, the sulfur is relayed by multiple sulfur mediators through persulfide chemistry⁴⁵, whereas in eukaryotes, sulfur transfer is facilitated by a ubiquitin-activating E1-like enzyme and ubiquitin-like proteins^{43,47}. In bacteria, loss of s²U modification due to deficiency of *mnmA* or sulfur mediators causes a slow growth phenotype⁴⁶, highlighting the physiological importance of this modification. Disruption of the CTU1/CTU2 complex impairs cytoplasmic s²U modification in yeast⁴⁸, worms⁴², and plants⁴⁹. Intriguingly, loss of cytoplasmic s²U showed a synthetic lethality together with deficiency of 5-methoxycarbonylmethyl modification of mcm⁵s²U⁴⁸. On the other hand, loss of mitochondrial s²U modification due to mutation of the yeast mitochondrial tRNA-specific 2-thiouridylase gene *mtu1* resulted in deficiency of mitochondrial protein synthesis and respiratory function⁵⁰. It is known that pathogenic mutations in the human *MTU1* gene cause reversible infantile liver failure (RILF), a heritable and acute liver failure followed by spontaneous recovery at an early stage of life^{51,52}. Loss-of-function mutations in the *CTU2* gene cause severe primary microcephaly, ambiguous male genitalia, dysmorphic facies, polydactyly, and renal agenesis (DREAM-PL)^{53,54}.

Due to the chemical properties of the thiocarbonyl group, xm⁵s²U modifications are chemically unstable, and readily oxidized and desulfurized under in vitro oxidative conditions, forming uridine or 4-pyrimidinone derivatives (xm⁵h²U) (Fig. 1b)^{55,56}. h²U predominantly adopts the C2'-*endo* form⁵⁷, suggesting less stable base pairing with A or G than s²U. Synthetic RNA duplexes containing an h²U • A pair

exhibited significantly lower melting temperatures than those with an s²U • A pair, while duplexes with either an s²U • G or h²U • G pair displayed similar melting temperatures⁵⁸. If desulfuration of xm⁵s²U occurs in cells, xm⁵h²U34 is likely to negatively regulate protein synthesis. In fact, mcm⁵h²U and cm⁵h²U, the desulfurized forms of mcm⁵s²U and cm⁵s²U, have been found in tRNAs from yeast and various human cultured cells regardless of the presence or absence of oxidative stress, reaching up to 11% depending on the efficiency of the antioxidant defense mechanisms⁵⁹. However, their mechanistic roles in translational regulation have not been elucidated.

Through a spike-in experiment, we confirmed that h²U derivatives are naturally occurring in cellular processes, rather than artifacts produced during tRNA extraction and purification. The mcm⁵h²U34 modification significantly impairs the aminoacylation efficiency of tRNAs for Lys, Glu, and Gln, eliminates AAA decoding, and substantially decreases AAG decoding at the A-site of the ribosome. This alteration affects protein synthesis in a human reconstituted in vitro translation system. Cryogenic electron microscopy (cryo-EM) structural analyses of ribosomes complexed with desulfurized tRNAs and AAR codons showed that the mcm⁵h²U • G pair has a minor groove-shifted U • G wobble geometry, whereas the mcm⁵h²U • A pair adopts a unique geometry with a single hydrogen bond (H-bond). These findings indicate that oxidative desulfuration of tRNA modifications influences protein synthesis under oxidative stress conditions.

Results

Detection of h²U derivatives in mammalian and bacterial tRNAs

We reviewed our previous liquid chromatography-mass spectrometry (LC-MS) data on cytoplasmic and mitochondrial tRNAs in mammalian tissues and human culture cells²², and detected unassigned nucleosides with a molecular mass 32 Da lower than that of mcm⁵s²U (*m/z* 333) and tm⁵s²U (*m/z* 398), implying that h²U derivatives are indeed present in cellular tRNAs. In LC-MS nucleoside analyses of total RNA from mouse livers, we clearly detected unassigned nucleosides with *m/z* 301 (Fig. 1c). In addition, we also detected nucleosides with *m/z* 366 in mitochondrial tRNA^{Lys} isolated from pig liver (Fig. 1d). Collision-induced dissociation (CID) of these nucleosides generated base-related ions (BH₂⁺) with loss of unmodified ribose (Fig. 1e, f), suggesting that the 32 Da loss originates from their bases.

To confirm that these nucleosides are h²U derivatives, we prepared authentic h²U derivatives by in vitro oxidation of mouse tRNA fraction and pig mitochondrial tRNA^{Lys}, and found massive accumulation of mcm⁵h²U (Fig. 1c) and tm⁵h²U (Fig. 1d). Next, we conducted LC-MS co-injection analyses of nucleosides from total RNA and its in vitro oxidized form, and demonstrated that the unassigned nucleosides co-eluted with the authentic h²U derivatives (Fig. 1c, d), confirming the presence of mcm⁵h²U and tm⁵h²U in mammalian tissues.

We also detected an unassigned nucleoside (*m/z* 272) with molecular mass 32 Da lower than that of mnm⁵s²U (*m/z* 304) in *Escherichia coli* tRNAs (Supplementary Fig. 1a, b). LC-MS co-injection with in vitro oxidized total tRNA confirmed the identity of this nucleoside as mnm⁵h²U (Supplementary Fig. 1a). To test whether mnm⁵h²U arises from desulfuration of mnm⁵s²U, we analyzed nucleosides from the *E. coli* Δ*mnmA* strain, which lacks 2-thiolation. In this strain, neither mnm⁵s²U nor mnm⁵h²U was detected (Supplementary Fig. 1c), supporting that mnm⁵h²U is derived from mnm⁵s²U. Similarly, we analyzed cytoplasmic tRNA^{Gln} from wild-type *Saccharomyces cerevisiae* and a *ctu1Δ* strain lacking 2-thiolation of mcm⁵s²U (Supplementary Fig. 1d). In wild-type tRNA^{Gln}, mcm⁵s²U was detected as the major modification, whereas the *ctu1Δ* strain showed only mcm⁵U and no detectable mcm⁵h²U (Supplementary Fig. 1d). These results confirm that mcm⁵h²U, like mnm⁵h²U, is generated from the corresponding 2-thiouridine derivative.

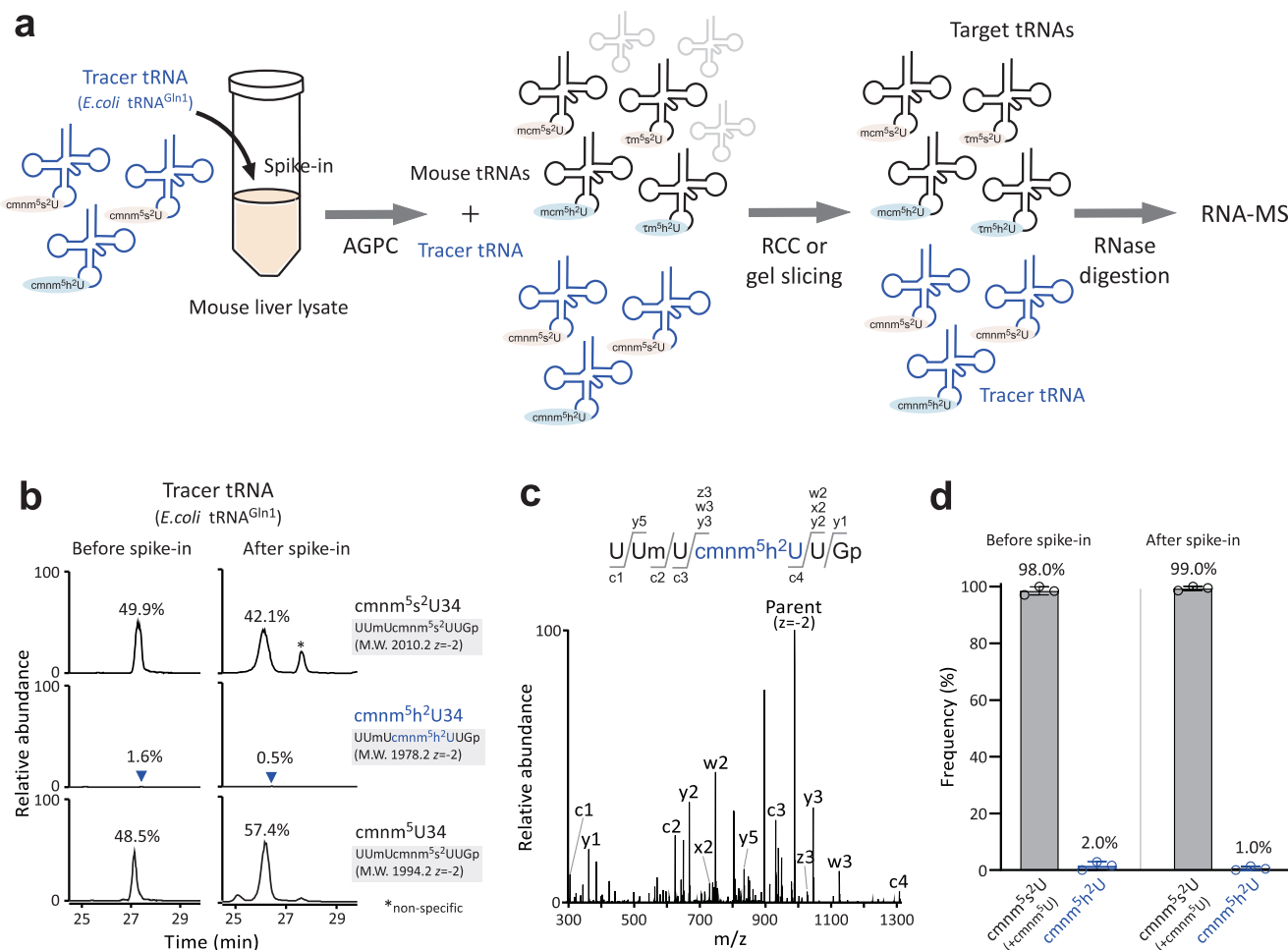


Fig. 2 | Spike-in experiment with tracer tRNA. **a** Scheme of the spike-in experiment with *E. coli* tRNA^{Gln1} as a tracer. **b** Status of tRNA modifications in the tracer tRNA before and after spike-in experiment. Extracted ion chromatograms (XICs) show the indicated negative ions of RNase T₁-digested fragments containing modified U34. Sequences of detected fragments, with their molecular weights and charge states, are displayed on the right. The relative modification frequencies (%)

indicated in chromatograms were calculated from the peak height ratios of RNA fragments with different modifications. A non-specific peak is asterisked. **c** CID spectrum of the cmnm⁵-h²U-containing fragment of *E. coli* tRNA^{Gln1}. **d** Frequencies of cmnm⁵-h²U and other uridine derivatives before and after spike-in experiments with technical triplicates. Results are means ± standard deviation. Source data are provided as a Source data file.

Confirmation of h²U derivatives produced in cells

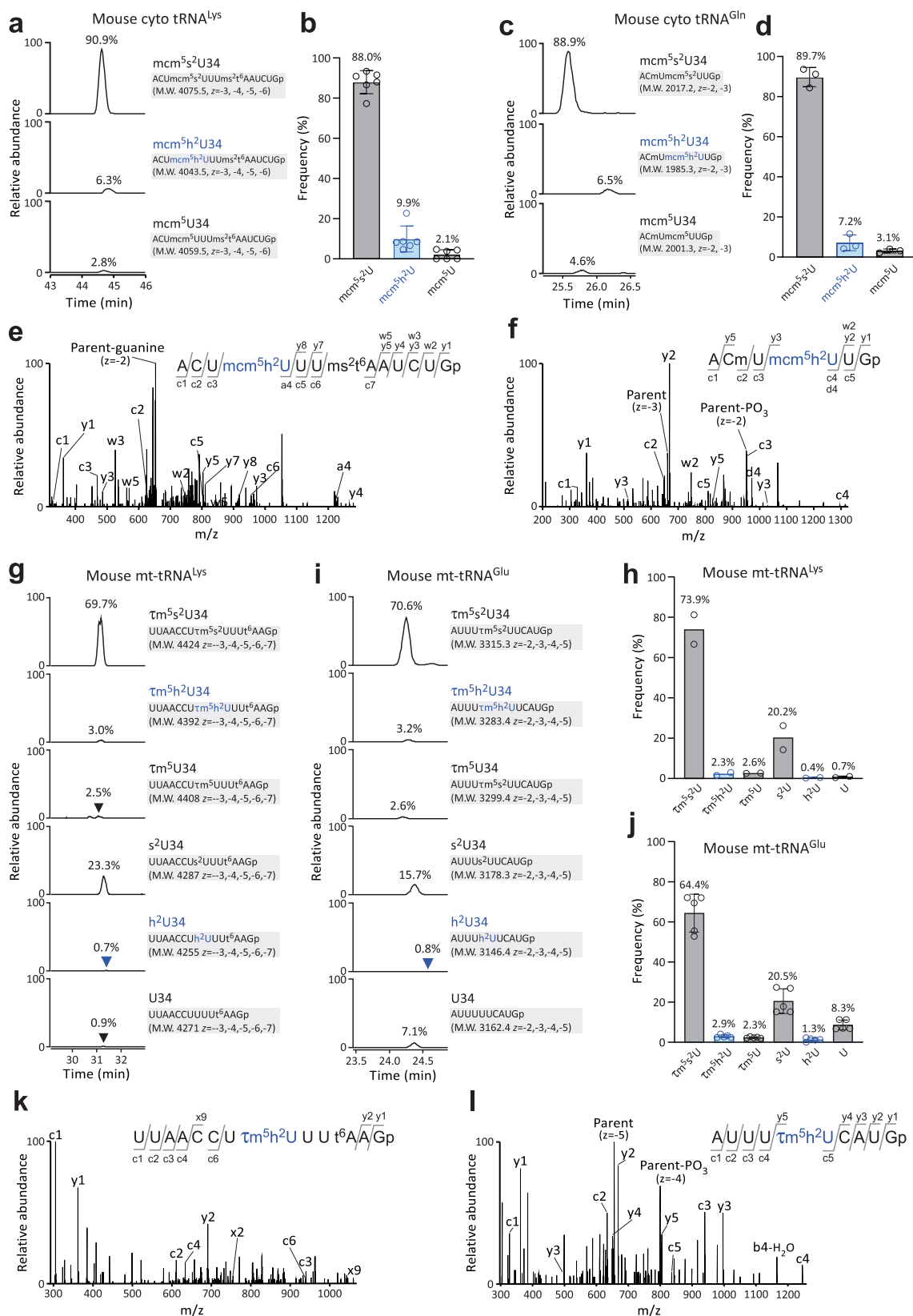
Given the chemical nature of xm⁵-s²U modifications, we initially hypothesized that their h²U derivatives might result from artificial oxidation during preparation of tRNAs in vitro. To confirm that h²U derivatives, namely, mcm⁵-h²U, tm⁵-h²U, and h²U, are indeed generated in mammalian cells, we conducted a spike-in experiment using *E. coli* tRNA as a tracer. This approach allowed us to estimate the extent of artificial oxidation of xm⁵-s²U during in vitro handling of tRNAs. As a tracer, we isolated *E. coli* tRNA^{Gln1}, which bears cmnm⁵-s²U at position 34 of the anticodon (Fig. 2a). This tRNA was chosen because cmnm⁵-s²U and its desulfurized derivative, cmnm⁵-h²U, are distinct from xm⁵-s²U and xm⁵-h²U found in mammalian tRNAs. In our experiment, *E. coli* tRNA^{Gln1} was added to a mouse liver homogenate as a spike-in tracer, followed by extraction of total RNA and purification of individual tRNAs, including the tracer, using reciprocal circulating chromatography (RCC)⁶⁰ (Fig. 2a).

To monitor artificial desulfuration of s²U derivatives during RNA extraction and tRNA purification, we measured the rate of conversion of cmnm⁵-s²U to cmnm⁵-h²U in the tracer tRNA (Fig. 2a). We isolated cytoplasmic tRNA^{Lys} and tRNA^{Glu}, as well as mitochondrial tRNA^{Lys} and tRNA^{Glu}, along with *E. coli* tRNA^{Gln1} as a tracer from mouse liver lysates. The tracer tRNA was then digested with RNase T₁, and its anticodon-

containing fragments were analyzed by capillary LC nano-electrospray ionization (ESI)-MS, revealing the presence of cmnm⁵-s²U, cmnm⁵-h²U, and cmnm⁵U (Fig. 2b). We detected cmnm⁵-h²U in ~2% of the tracer tRNA prior to the spike-in experiment (Fig. 2b and Supplementary Fig. 2a). CID analysis confirmed the presence of cmnm⁵-h²U at position 34 (Fig. 2c). After isolation with mouse tRNAs, the level of cmnm⁵-h²U remained virtually unchanged (Fig. 2b, d, and Supplementary Fig. 2b, c), indicating that artificial formation of h²U derivatives during handling of tRNAs was negligible under our experimental conditions.

Using this approach, we identified mcm⁵-h²U in 9.9% and 7.2% of mouse cytoplasmic tRNA^{Lys} and tRNA^{Gln}, respectively (Fig. 3a–d). CID analyses confirmed the presence of mcm⁵-h²U at position 34 in both tRNAs (Fig. 3e, f). Similarly, tm⁵-h²U was detected in 2.3% and 2.9% of mitochondrial tRNA^{Lys} and tRNA^{Glu}, respectively (Fig. 3g–j). h²U was also identified in 0.4% and 1.3% of these tRNAs, respectively (Fig. 3g–j). The presence of tm⁵-h²U at position 34 in both mitochondrial tRNAs was further validated by CID analyses (Fig. 3k, l).

To expand on these findings, we performed similar analyses in various cultured cell lines (Supplementary Fig. 3). We observed that mcm⁵-h²U was present at varying levels across different cell lines. Notably, ~15% of tRNA^{Lys} in HeLa cells contained mcm⁵-h²U, while it was scarcely detected in A549 and T24 cells.



Translation efficiency decreases upon desulfuration of tRNAs

Given that 2-thiouridine derivatives play crucial roles in accurate and efficient translation, we wondered whether h²U derivatives affect translation in a codon-specific manner. We examined the function of desulfurized tRNAs using a human cell-free translation system. To this end, we prepared desulfurized tRNAs by in vitro oxidation with H₂O₂⁵⁶

and confirmed efficient conversion of mcm^sU to mcm^hU (Supplementary Fig. 4a). As a reporter mRNA, *Renilla* luciferase (Rluc) with an N-terminal hemagglutinin (HA) was used (Fig. 4a), and translation initiation was mediated by the Hepatitis C virus (HCV) internal ribosome entry site (IRES) (Fig. 4a). As a control mRNA, we employed a mutant HiBiT peptide that is unaffected by tRNA desulfuration

Fig. 3 | Detection of h²U derivatives in cytoplasmic and mitochondrial tRNAs. Status of wobble modifications in mouse cytoplasmic tRNAs for Lys (a) and Gln (c), and mitochondrial tRNAs for Lys (g) and Glu (i). XICs show the indicated negative ions of RNase T₁-digested fragments containing modified U34. Sequences of detected fragments, with their molecular weights and charge states, are displayed on the right. The relative modification frequencies (%) indicated in chromatograms were calculated from the peak height ratios of RNA fragments with different

modifications. Frequencies of h²U derivatives and other uridine derivatives with technical replicates of mouse cytoplasmic tRNAs for Lys (b, *n* = 6) and Gln (d, *n* = 3), and mitochondrial tRNAs for Lys (h, *n* = 2) and Glu (j, *n* = 5). Results are means ± standard deviation. Source data are provided as a Source data file. CID spectra of h²U-containing fragments of mouse cytoplasmic tRNAs for Lys (e) and Gln (f), and mitochondrial tRNAs for Lys (k) and Glu (l).

(Fig. 4a). Since mcm⁵s²U is present in four mammalian cytoplasmic tRNAs for Lys, Glu, Gln, and Arg (encoded by AGR codons), two Lys residues in the HiBiT peptide (VSGWRLFKKIS) were substituted with Arg encoded by CGN codons to generate a mutant HiBiT construct (SGWRLFRRIIS) which forms a complex with Large BiT, reconstituting functional nano luciferase activity.

The Rluc reporter mRNA and either desulfurized or untreated total tRNA were added to a reconstituted human cell-free protein synthesis system consisting of human 80S ribosomes, translation factors, and aminoacyl-tRNA synthetases⁶¹. The activity of Rluc was measured by luciferase assay. As expected, significant Rluc activity was observed when untreated tRNA was used, whereas it was markedly reduced when desulfurized tRNA was used (Fig. 4b). This result was corroborated by western blotting using an anti-HA antibody (Fig. 4c). To demonstrate that this effect is specific to certain codons, the mutant HiBiT mRNA was translated and nano luciferase activity was measured. We observed similar levels of activity regardless of whether desulfurized or untreated tRNA was used (Fig. 4b). Among 378 amino acid residues in the Rluc reporter sequence, 77 codons are read by the four tRNA species containing mcm⁵s²U, accounting for 22% of the total (Supplementary Data 1). Specifically, it consists of 62 A-ending codons and 15 G-ending codons. The G-ending codons are redundantly decoded by isodecoders having either mcm⁵s²U34 or C34, whereas the A-ending codons are decoded solely by mcm⁵s²U34-tRNAs. These findings suggest that the translational activity of mcm⁵s²U-modified tRNAs is impaired by desulfuration.

Since desulfurized tRNAs had less activity in an in vitro translation, we next examined whether desulfurized tRNAs participate in protein synthesis in cells by detecting mcm⁵h²U in polysome-associated tRNAs. HEK293T cell lysate was first separated into the supernatant (free RNA) and crude ribosome fractions using a sucrose cushion (Fig. 4d), followed by sucrose density gradient (SDG) centrifugation to obtain the polysome fraction (Fig. 4e). Small RNA were then obtained from the free RNA and polysomal RNA, and subjected to the LC-MS nucleoside analysis, showing that mcm⁵h²U is present in the polysomal RNA (Fig. 4f, g). As expected, the signal intensity of mcm⁵h²U in the polysomal RNA was lower than that in the total RNA and free RNA fractions. These results suggest that a portion of desulfurized tRNAs participates in translation, whereas others might be excluded.

Effect of tRNA desulfuration on aminoacylation

It has been revealed that the desulfuration of mcm⁵s²U impairs protein synthesis. We next investigated which step of protein synthesis is affected by mcm⁵h²U. Although 2-thio modification of xm⁵s²U is known to contribute to efficient aminoacylation of tRNAs in bacteria^{31–33}, the role of the 2-thio group of mcm⁵s²U in tRNA aminoacylation remains poorly understood in mammals. To examine the effect of desulfuration of mcm⁵s²U on aminoacylation, we performed in vitro aminoacylation reactions using recombinant human aminoacyl-tRNA synthetases and total tRNAs with or without desulfuration.

As a control, we used tRNAs^{Thr}, which do not have mcm⁵s²U. Little change in Thr incorporation in total tRNAs was observed regardless of desulfuration treatment (Fig. 5a). Next, we compared the incorporation activities of four amino acids (Arg, Gln, Glu, and Lys) into total tRNAs with or without desulfuration. For Arg, no difference in the

incorporation efficiency was observed between desulfurized and untreated total tRNA (Fig. 5b), indicating that arginyl-tRNA synthetase (RARS1) does not recognize mcm⁵s²U34 in tRNA^{Arg(AGR)} for its aminoacylation. By contrast, significant reductions in aminoacylation activity were observed for tRNAs for Gln, Glu, and Lys upon desulfuration (Fig. 5c–e). Considering that tRNAs charged with these amino acids are a mixture of isoacceptors including isodecoders (Supplementary Data 2)⁶², some parts of which carry mcm⁵s²U34, it became evident that desulfuration of mcm⁵s²U34 significantly impairs aminoacylation of these tRNAs. Thus, we found that the three aminoacyl-tRNA synthetases, glutaminyl-tRNA synthetase (QARS1), glutamyl-prolyl-tRNA synthetase 1 (EPRS1), and lysyl-tRNA synthetase (KARS1), specifically recognize the chemical structure of the 2-thiocarbonyl group in mcm⁵s²U to achieve efficient aminoacylation. These findings indicate that desulfuration of mcm⁵s²U negatively regulates cytoplasmic translation by lowering the aminoacylation efficiency of tRNAs for Gln, Glu, and Lys.

Effect of tRNA desulfuration on codon recognition

Next, we investigated the effect of mcm⁵s²U desulfuration on tRNA codon recognition on the ribosome. The xm⁵s²U34 modification is crucial for accurate and efficient recognition of purine nucleosides at the third position of NNR codons^{17,34}. We measured the ability of human tRNA^{Lys} with or without desulfuration to bind both AAA and AAG codons at the A-site on the ribosome. As a positive control, untreated tRNA^{Lys} efficiently bound to both codons (Fig. 6a, b). Desulfurized tRNA^{Lys} showed almost no binding to the AAA codon (Fig. 6a) and a ~60% reduction in binding to the AAG codon (Fig. 6b). These results reveal that mcm⁵h²U34 significantly impairs codon recognition at the A-site on the ribosome.

Structural insights into codon recognition mediated by h²U derivatives

To obtain structural insights into the molecular mechanism by which mcm⁵h²U34 reduces codon recognition ability, we conducted cryo-EM single-particle analyses of the ribosome complex with human cytoplasmic tRNA^{Lys} bearing mcm⁵s²U34 or mcm⁵h²U34, P-site tRNA, and mRNA with AAA or AAG at the A-site. To this end, human cytoplasmic tRNA^{Lys} was desulfurized by treating twice with H₂O₂, confirming the high frequency of mcm⁵h²U34 (Supplementary Fig. 4b). After several classification and focused classification targeting A-site tRNA, we determined cryo-EM structures of four ribosome complexes with A- and P-site tRNAs at 2.26–3.18 Å resolution (Fig. 7a, Supplementary Figs. 5a–d, Supplementary Fig. 6, and Supplementary Data 3). Ribosomes adopt the closed conformation in all four structures (Fig. 7b), with the codon-anticodon helix at the A-site strictly recognized by A-minor interactions with A1492 and A1493 in helix 44 and G530 in helix 18 of 16S ribosomal RNA (rRNA). In particular, we clearly observed cryo-EM densities for 5-methoxycarbonylmethyl- and 2-thiocarbonyl groups of mcm⁵s²U34 (Fig. 7c). On the other hand, no density was observed for the 2-thiocarbonyl group in mcm⁵h²U34 (Fig. 7d). Additionally, the 5-methoxycarbonylmethyl groups of mcm⁵s²U34 and mcm⁵h²U34 interact with C1054 of 16S rRNA in all four structures (Supplementary Fig. 7a–d).

The mcm⁵s²U34 modification pairs with A3 of the AAA codon via a canonical Watson-Crick geometry (Fig. 7c), whereas mcm⁵s²U34 pairs

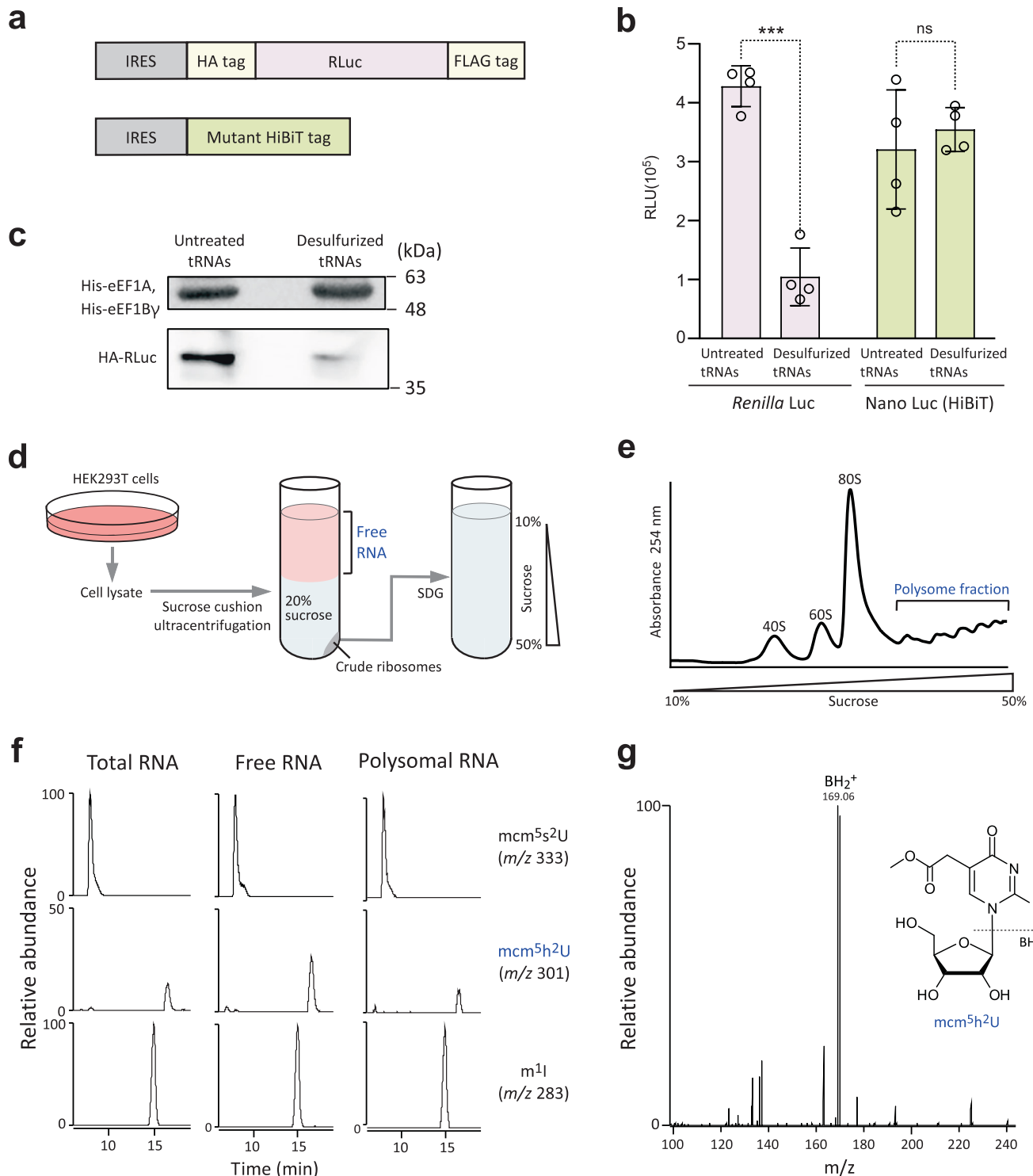


Fig. 4 | Desulfuration of tRNA modification affects translation efficiency.

a Schematic diagram of HCV-IRES mRNAs of RLuc (upper picture) and mutant HiBiT (lower picture) for human cell-free translation. **b** Luciferase activities of RLuc (left bars) and mutant HiBiT (right bars) translated in vitro with either desulfurized or untreated total tRNA. Results are means $\pm$ standard deviation from four independent experiments. *** $p < 0.001$ ($p = 0.00007$, 95% CI = 248051–399017, $t(5.39) = 10.78$, Cohen's $d = 7.62$). n.s., not significant ($p = 0.566$, 95% CI = –186481–118851, $t(3.80) = -0.63$, Cohen's $d = -0.44$). Unpaired two-tailed Welch's t -test. Source data are provided as a Source data file. **c** Western blotting of HA-tagged RLuc with an anti-HA antibody. His-tagged eEF1A and eEF1B γ were

blotted with an anti-His antibody as loading controls. Source data are provided as a Source data file. **d** Scheme for polysome fraction preparation from HEK293T cells. The free RNA fraction in the supernatant of the sucrose cushion is indicated. **e** SDG profiling of crude ribosome from HEK293T cells. The polysome fraction is indicated. **f** LC-MS nucleoside analyses of total RNA (left panels), free RNA (middle panels), and polysomal RNA (right panels) from HEK293T cells. Mass chromatograms represent proton adducts of mcm⁵s²U (m/z 333, upper panels), mcm⁵h²U (m/z 301, middle panels), and m¹I (m/z 283, lower panels). The relative peak intensity is normalized against mcm⁵s²U intensity in each analysis. **g** CID spectrum of mcm⁵h²U in polysomal RNA. The base-related ion (BH₂⁺) is indicated.

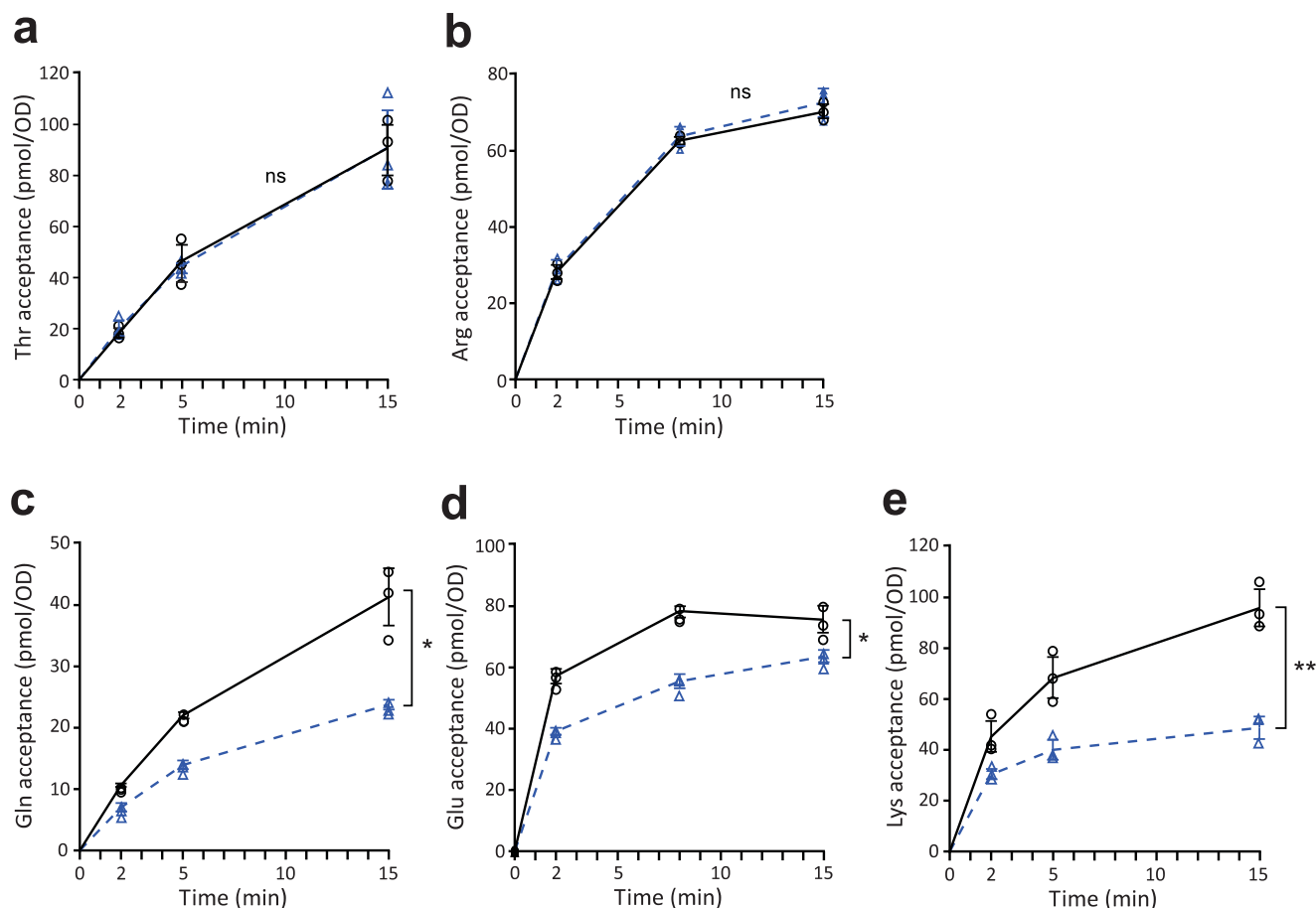


Fig. 5 | Desulfuration of tRNA modification affects aminoacylation. Aminoacylation of desulfurized (blue dotted lines) and untreated (black lines) total tRNAs for Thr (**a**), Arg (**b**), Gln (**c**), Glu (**d**) and Lys (**e**). Results are means $\pm$ standard deviation from three independent experiments. * $p < 0.05$ (**c**: $p = 0.0337$, 95% CI = 3.20–31.83, $t(2.11) = 5.00$, Cohen's $d = 4.08$; **d**: $p = 0.0486$, 95% CI = 0.14–23.84, $t(2.83) = 3.34$,

Cohen's $d = 2.72$). ** $p < 0.01$ (**e**: $p = 0.0035$, 95% CI = 28.13–66.32, $t(3.30) = 7.49$, Cohen's $d = 6.11$). n.s., not significant (**a**: $p = 0.9877$, 95% CI = –42.01–41.58, $t(3.4) = -0.02$, Cohen's $d = -0.01$; **b**: $p = 0.4668$, 95% CI = –12.06–7.07, $t(3.03) = -0.83$, Cohen's $d = -0.68$). Unpaired two-tailed Welch's t -test. Source data are provided as a Source data file.

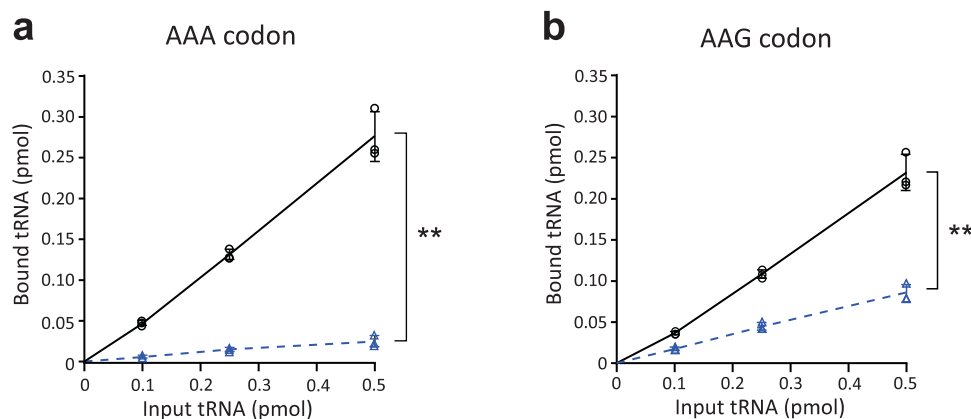


Fig. 6 | Desulfuration of tRNA^{lys} affects codon recognition. Ribosomal A-site tRNA-binding experiments performed for AAA (**a**) and AAG (**b**) codons with desulfurized (blue dotted lines) and untreated (black lines) tRNA^{lys}. Data were normalized against non-specific binding to the UUC codon. Results are means $\pm$

standard deviation from three independent experiments. ** $p < 0.01$ (**a**: $p = 0.0048$, 95% CI = 0.1686–0.3355, $t(2.20) = 11.91$, Cohen's $d = 9.72$; **b**: $p = 0.0037$, 95% CI = 0.0914–0.1999, $t(2.89) = 8.73$, Cohen's $d = 7.13$). Unpaired two-tailed Welch's t -test. Source data are provided as a Source data file.

with G3 of the AAG codon via a non-canonical base pairing involving a minor groove-shifted U (Fig. 7c). The geometry of the mcm⁵s²U • G3 base pair is discussed in more detail in the Supplementary Discussion. Meanwhile, mcm⁵h²U34 pairs with A3 via a unique geometry involving a single H-bond between O4 of mcm⁵h²U34 and N6 of A3 (Fig. 7d, e).

The N3 atom of mcm⁵h²U no longer acts as a proton donor due to its chemical nature, so that it is not involved in any additional H-bonding (Fig. 7e). The observed geometry with a single H-bond could explain why s²U-to-h²U conversion results in loss of the ability to recognize the AAA codon (Fig. 6a).

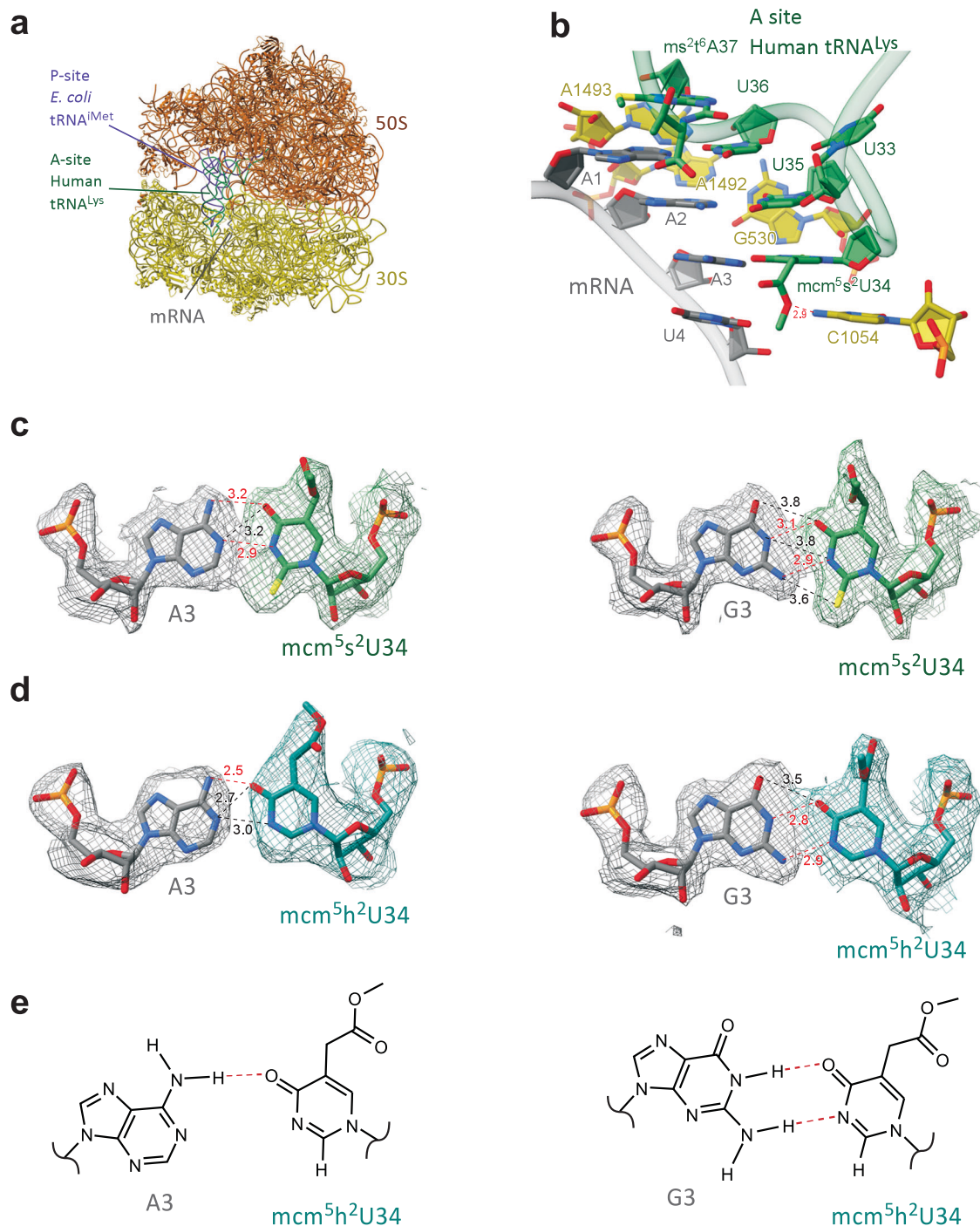


Fig. 7 | Structural basis of AAR decoding by mcm⁵s²U34 and mcm⁵h²U34 of tRNA^{Lys}. **a** Atomic model of the *E. coli* 70S ribosome bound to *E. coli* tRNA^{iMet} (purple) at the P-site and human cytoplasmic tRNA^{Lys} (green) at the A-site. **b** Codon-anticodon duplex of human cytoplasmic tRNA^{Lys} (green) and the AAA codon (gray) in the decoding center of 16S rRNA (lime yellow). Base pairing

geometries of mcm⁵s²U34 (c) and mcm⁵h²U34 (d) recognizing A3 of the AAA codon (left panels) and G3 of the AAG codon (right panels). Distances between key atoms involved in the interaction are indicated. **e** Chemical structures of base pairing between mcm⁵h²U and A3 (left panel) or G3 (right panel).

The mcm⁵h²U • G pair adopts a similar geometry to the non-canonical wobble base pair mmm⁵s²U • G, with U* shifted into the minor groove (Fig. 7d, e). Given that the 2-thiocarbonyl group of mcm⁵s²U stabilizes C3' *endo* ribose pucker^{36,37}, mcm⁵s²U • G pairing should be more stable than mcm⁵h²U • G pairing. In addition, the 2-thiocarbonyl group of mcm⁵s²U lowers the pKa of N3 atom⁴⁰, and the 5-methylcarboxymethyl group also contributes to the structural rearrangement to form the minor groove-shifted mcm⁵s²U • G pairing³⁸.

This structural observation is consistent with our biochemical evidence showing that s²U-to-h²U conversion decreases the ability to decode the AAG codon (Fig. 6b).

Discussion

Through a spike-in experiment, we demonstrated that 2-thiouridine derivatives in tRNA are desulfurized in cells and converted to h²U derivatives, but these are relatively scarce tRNA modifications; in

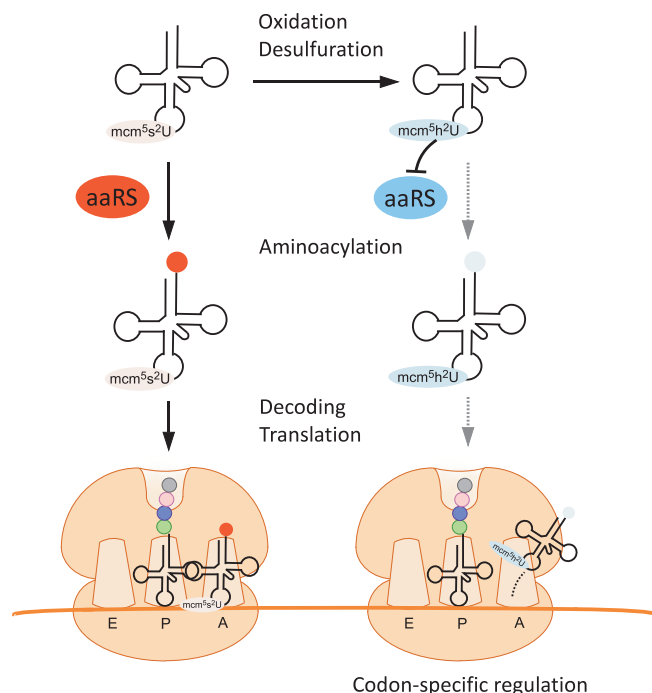


Fig. 8 | Proposed mechanism of translational regulation mediated by desulfurized tRNA modification. Under oxidative stress conditions, mcm⁵s²U34 modification of cytoplasmic tRNAs for Lys, Gln, Glu, and Arg undergoes desulfuration, yielding mcm⁵h²U34. mcm⁵h²U34 decreases the aminoacylation efficiency of tRNA^{Lys}, tRNA^{Gln}, and tRNA^{Glu}, and also impairs codon recognition at the A-site on the ribosome. As a result, codon-specific translation efficiency is negatively regulated.

mouse tissues, the frequency of mcm⁵h²U was <10% of cytoplasmic tRNAs bearing mcm⁵s²U (Fig. 3b, d), and tm⁵h²U was detected in only a few percent of mitochondrial tRNAs with tm⁵s²U (Fig. 3h, j). However, the abundance of h²U derivatives varies in different cell types. Approximately 15% of tRNA^{Lys} in HeLa cells contained mcm⁵h²U, whereas little was detected in A549 and T24 cells (Supplementary Fig. 3). Since h²U derivatives can be generated by oxidative desulfuration in vitro^{55,56}, it is highly likely that they are chemically produced in cells by reactive oxygen species generated under oxidative stress conditions. Nevertheless, enzymatic formation of h²U derivatives cannot be ruled out. In vitro oxidative desulfuration of s²U derivatives yields both h²U and uridine derivatives; however, since the desulfuration mechanism in cells remains unclear, the ratio in which these two types of products are formed is unknown. While h²U derivatives are likely to represent a form of damaged RNA, the uridine derivatives generated by desulfuration may be re-thiolated back to 2-thiouridine, suggesting that this is a reversible modification. Understanding the mechanisms underlying the differing frequencies of h²U in various cells and tissues is intriguing. The frequency of h²U is likely determined by multiple factors, including the cellular oxidative stress response, redox balance, and the degradation of h²U-modified tRNAs. Because s²U derivatives play a crucial role in codon-specific translation, our discovery suggests that tRNA s²U modification serves as a sensor to monitor intracellular oxidative states to regulate codon-specific translation (Fig. 8). In fact, although we quantified h²U derivatives under various oxidative stress conditions, we have not yet identified any conditions that reproducibly increase h²U levels (Supplementary Data 4). Sierant et al.⁵⁹ likewise detected mcm⁵h²U in yeast and human cultured cells and analyzed its behavior under oxidative stress using calibration curves derived from synthetic standards; however, their results also showed that the levels of h²U species do not markedly increase upon exogenous oxidative stress. Cells counteract reactive oxygen species (ROS) primarily through enzymatic antioxidant

systems, such as superoxide dismutase, catalase, and glutathione peroxidase⁶³. At supraphysiological concentrations of H₂O₂, however, these enzymatic ROS-scavenging systems are rapidly overwhelmed, leading to cell death. Under these conditions, the limited impact of exogenous oxidative stress on h²U levels observed in our experiments (Supplementary Data 4) is more plausibly explained by the chemical buffering capacity of the intracellular environment, including abundant proteins, metabolites, and other redox-active components, rather than by regulated enzymatic antioxidant defenses^{64,65}. The capacity and contribution of these mechanisms vary substantially among cell types, and this variation might be reflected in the different h²U levels observed across the cell lines examined (Supplementary Fig. 3).

Cell-free protein synthesis experiments using human translational apparatus reconstituted in vitro revealed that h²U derivatives suppress codon-specific translation (Fig. 4b, c). Specifically, h²U derivatives decreased the aminoacylation efficiency of cytoplasmic tRNAs for Lys, Gln, and Glu, but not for Arg (Fig. 5b–e). We also showed that tm⁵h²U and h²U are present in mitochondrial tRNAs (Fig. 3h, j), and it is plausible that they also suppress aminoacylation of mitochondrial tRNAs. Furthermore, in the codon recognition of cytoplasmic tRNA^{Lys}, mcm⁵h²U almost abolished the ability to recognize the AAA codon and significantly reduced recognition of the AAG codon (Fig. 6a, b). Similar effects can be predicted for other cytoplasmic and mitochondrial tRNAs containing h²U derivatives, although the degree of suppression likely varies depending on the codons. Additionally, given that s²U modifications in bacterial tRNAs are involved in efficient translocation³⁵, it is probable that h²U derivatives regulate translocation efficiency on mammalian ribosomes. Mass spectrometric analyses of the polysomal RNA clearly detected mcm⁵h²U (Fig. 4f, g), but its level was lower than that in the total RNA and free RNA fractions, indicating that a part of desulfurized tRNAs participates in translation, whereas the others do not due to their low abilities of aminoacylation and decoding.

Cryo-EM analysis of ribosome complexes revealed the structural basis for codon recognition by mcm⁵h²U34 at the A-site of the ribosome (Fig. 7d, e). While the base pairing geometry of the mcm⁵h²U34 • A3 pair resembles the Watson-Crick type, it involves only a single H-bond (Fig. 7e), indicating a very weak interaction. This result is consistent with previous findings on the thermodynamic stability of double-stranded RNAs containing h²U, as well as with their quantum chemical calculations⁵⁸.

By contrast, the mcm⁵h²U34 • G3 pair adopts a geometry similar to the minor groove-shifted U • G wobble pair observed in the mcm⁵s²U34 • G3 pair, with the uracil base shifted toward the minor groove. This geometry arises because the N3 atom of mcm⁵h²U34 functions as a proton acceptor for the 2-amino group of G3 (Fig. 7e). However, unlike the mcm⁵s²U34 • G3 pair (Fig. 7c), the mcm⁵h²U34 • G3 pair lacks the stabilization provided by the 2-thiocarbonyl group (Fig. 7d, e), which enhances stacking effect with a neighboring base pair. As a result, the mcm⁵h²U34 • G3 pair is relatively weaker.

Building on these findings, we propose a hypothetical mechanism for translation regulation mediated by desulfurized tRNA modifications (Fig. 8). Under oxidative stress conditions, the mcm⁵s²U34 modification of cytoplasmic tRNAs for Lys, Gln, Glu, and Arg undergoes desulfuration, yielding mcm⁵h²U34 or mcm⁵U34. This leads to reduced aminoacylation efficiency for tRNA^{Lys}, tRNA^{Gln}, and tRNA^{Glu}, as well as impaired codon recognition at the ribosomal A-site. As a result, codon-specific translation efficiency is negatively impacted. Optimal translation speed is crucial for maintaining mRNA stability^{66–69} and ensuring correct folding of nascent proteins⁷⁰. Therefore, desulfuration of xm⁵s²U modifications may cause codon-specific ribosome stalling, potentially triggering pathways such as codon optimality-mediated decay⁶⁶ or the unfolded protein response⁷¹. Additionally, ribosome stalling during translation elongation could lead to ribosome collisions, activating the ribosome quality control pathway and inducing the integrated stress response or ribotoxic stress response^{72–74}.

In this study, we demonstrated that tRNAs bearing h²U derivatives indeed participate in protein synthesis and that they markedly impair aminoacylation efficiency and codon recognition ability. Based on these findings, we performed a relative synonymous codon usage (RSCU) analysis (Supplementary Data 5) on genes induced under oxidative stress conditions (GO:0006979). Compared with the RSCU values calculated from all human protein-coding genes⁷⁵, we observed no significant differences in codon usage for Lys, Gln, or Glu. However, for Arg, the usage of CGN codons was decreased, while that of AGR codons was relatively increased. Desulfuration of mcm⁵s²U led to reduced aminoacylation capacity in tRNAs for Lys, Gln, and Glu, whereas tRNA^{Arg(AGR)} retained full aminoacylation activity. Although the codon recognition properties of tRNA^{Arg(AGR)} bearing mcm⁵h²U remain to be investigated, its resistance to desulfuration may contribute to the preferential usage of AGR codons in oxidative stress-responsive genes.

In future work, our goal is to identify oxidative stress conditions that effectively promote the accumulation of h²U derivatives. This will allow us to further investigate the biological significance of desulfurized tRNA modifications in specific biological contexts.

Methods

Biological resources

E. coli strain BW25113 was obtained from National BioResource Project *E. coli* (NBRP-*E. coli*) at the National Institute of Genetics (Mishima, Japan). The Δ *mnmA* strain was obtained from the *E. coli* KEIO collection resource (Tokyo, Japan)⁷⁶. *S. cerevisiae* BY4742 (WT) strain (*Mat* α ; *his3-1*; *leu2-0*; *ura3-0*) and *ctt1* Δ strain (BY4742 background) were obtained from EUROSCARF (Germany). Mouse and pig livers were purchased from Funakoshi (Tokyo, Japan). MKN45 (RCB1001), HeLa (RCB0007), HEK293T (RCB2202) and HuH-7 (RCB1366) were obtained from RIKEN BRC (Tsukuba, Japan). A549 (CCL-185), Hep G2 (HB-8065), MCF7 (HTB-22), T24 (HTB-4), and RAW 264.7 (TIB-71) were obtained from ATCC. MOLM-13 (C0003003) was obtained from Addexbio (San Diego).

Cell culture

E. coli cells were cultivated in Luria-Bertani (LB) medium with vigorous shaking at 37 °C. *S. cerevisiae* cells were cultivated in Yeast Peptone Dextrose (YPD) medium (2% peptone, 1% yeast extract, and 2% glucose) at 30 °C. HeLa, HuH-7, hepG2, MCF7, and A549 cells were cultured at 37 °C and 5% CO₂ in Dulbecco's modified Eagle's medium (high-glucose; Sigma) supplemented with 10% fetal bovine serum (FBS; Gibco) and 1% penicillin-streptomycin (Fujifilm Wako Pure Chemical Corporation). MOLM-13, MKN45, and RAW 264.7 cells were cultured at 37 °C and 5% CO₂ in Roswell Park Memorial Institute (RPMI) 1640 medium (Nacalai Tesque) supplemented with 10% FBS. T24 cells were cultured at 37 °C and 5% CO₂ in McCoy's 5A (modified) Medium (Gibco) supplemented with 10% FBS.

For oxidative stress conditions, HeLa cells were cultured until they reached 70–80% confluency, then treated for 1 h with oxidative stress-inducing agents: 2 mM H₂O₂ (Nacalai Tesque), 20 μ M menadione (Sigma-Aldrich), or 0.25–1 mM rotenone (Tokyo Chemical Industry). Although 2 mM H₂O₂ is far above the IC₅₀ value for HeLa cells (50 μ M), we did not observe any cell detachment nor apoptosis in our condition. Following treatment, cells were washed with PBS. Total RNA was extracted as shown below.

RNA extraction and isolation of individual tRNAs

Total RNA was extracted from *E. coli* as described⁷⁷. Briefly, *E. coli* cells were suspended in a 1:1 mixture of water-saturated phenol and RNA extraction buffer A [50 mM NaOAc (pH 5.2) and 10 mM Mg(OAc)₂], followed by two cycles of freeze-thawing with liquid nitrogen, and vigorous stirring for 1 h at room temperature. After centrifugation, the aqueous phase was subjected to chloroform extraction and 2-propanol precipitation.

Total RNA from yeast cells were prepared as follows. Yeast cells were resuspended in 5–10 weights of RNA extraction buffer B [50 mM NaOAc (pH 5.2), 10 mM EDTA, 1% SDS, and 14.3 mM β -mercaptoethanol], and an equal volume of water-saturated phenol was added. The mixture was shaken for 1 h, followed by centrifugation at 8000 $\times$ g for 30 min at 4 °C. The supernatants were collected, treated with an equal volume of chloroform, followed by isopropanol precipitation. The pellets were dissolved in water, and added with 2% (v/v) 1 M MES-NaOH (pH 6.0), 2.5% (v/v) 5 M NaCl, and 40% (v/v) 2-butoxyethanol. The mixture was then placed on ice for 30 min, followed by centrifugation at 8000 $\times$ g for 30 min at 4 °C to remove precipitates. The supernatants were subjected to ethanol precipitation to obtain the total RNA. To obtain small RNA fraction, total RNA was subjected to an anion-exchange resin (DEAE Sepharose Fast Flow; Cytiva) in low salt buffer A [20 mM HEPES-KOH (pH 7.5), 250 mM NaCl, and 2 mM dithiothreitol (DTT)]. After washing the resin with the same buffer, small RNA fractions were eluted with high salt buffer A [20 mM HEPES-KOH (pH 7.5), 1 M NaCl, and 2 mM DTT].

Total RNA from cultured cells or frozen tissue were prepared using an isolation reagent comprising 40% (w/v) phenol, 1% (v/v) ethylene glycol, 2 M guanidine thiocyanate, 20 mM tri-sodium citrate, 0.1 M NaOAc (pH 4.5), 0.2% (w/v) N-lauroylsarcosine sodium salt, 50 mM salicylic acid, 10 μ M aluminon, 6.9 mM 8-hydroxyquinoline, 0.0001 w/v% BPB. For highly reducing conditions, the same isolation reagent was used with the addition of 50 mM DTT. Alternatively, total RNA were extracted from cultured cells or frozen tissue using TriPure Isolation Reagent (Roche). Frozen tissues were cryogenically ground in a GRINDOMIX GM 200 knife mill (Roche) prior to extraction. For further purification, total RNA were subjected to an anion-exchange resin (DEAE Sepharose Fast Flow; Cytiva) in low salt buffer B [10 mM HEPES-KOH (pH 7.5), 250 mM NaCl, and 2 mM DTT], and small RNA fractions were eluted with high salt buffer B [10 mM HEPES-KOH (pH 7.5), 1 M NaCl and 2 mM DTT]. To obtain total tRNA fractions or class I tRNA fractions, total RNA was separated by on 7 M urea denaturing polyacrylamide gel electrophoresis (PAGE) and stained with SYBR Gold (Invitrogen), then purified by excising the corresponding gel bands. Individual tRNAs were isolated by RCC^{60,78}. Sequences of 5'-terminal ethylcarbamate (EC) amino-modified DNA probes (Sigma Aldrich) complementary to target tRNAs are listed in Supplementary Data 6. Each probe was covalently immobilized on NHS-activated Sepharose 4 Fast Flow resin (Cytiva), packed in a custom-made tip, and attached to a custom-made multichannel head on the RCC device. Small RNA fractions obtained by DEAE chromatography were dissolved in 6 $\times$ NHE buffer [1.2 M NaCl, 30 mM HEPES-KOH (pH 7.5), 15 mM EDTA-NaOH pH 8.0 and 1 mM DTT], and passed through the tip by auto-pipetting at 65 °C. After washing the tip columns with 0.1 $\times$ NHE at 40 °C, bound tRNAs were eluted with 0.1 $\times$ NHE at 68 °C. The purity of isolated tRNAs was confirmed by 10% PAGE with 7 M urea.

RNA mass spectrometry

Isolated tRNAs were digested with RNase T₁ and analyzed by capillary LC/nano-ESI-MS as described^{78–80}. One picomole of isolated tRNA was digested with 20 U of RNase T₁ (Thermo Fisher Scientific) in 20 mM NH₄OAc (pH 5.3) at 37 °C for 45 min. A linear ion trap-orbitrap hybrid MS instrument (LTQ Orbitrap XL; Thermo Fisher Scientific) was equipped with a splitless nano HPLC system (DiNa; Techno Alpha) using a nano-LC trap column (L-column2 ODS; 0.3 $\times$ 5 mm; CERI) and a capillary column (HiQ Sil C18W-3; 0.1 $\times$ 100 mm; Techno Alpha). Digested fragments were separated for 35 min at a flow rate of 300 nL/min with a linear gradient from 2 to 100% solvent B. The solvent system consisted of 0.4 M 1,1,1,3,3,3-hexafluoro-2-propanol (HFIP; pH 7.0, solvent A) and 0.4 M HFIP (pH 7.0) in 50% methanol (solvent B). The eluent was ionized using an ESI source in negative mode, with a scan range of *m/z* 600–2000. Xcalibur 2.0.7 (Thermo Fisher Scientific) was used for system operation.

LC-MS shotgun analyses of tRNA modification were conducted as described⁸¹ with slight modification. Five picomoles of class I tRNA fraction was digested with 50 units of RNase T₁ (Thermo Fisher Scientific) in 20 mM NH₄OAc (pH 5.3) at 37 °C for 45 min. Digested fragments were analyzed by a DiNa nano-flow HPLC system (Techno Alpha) coupled with an Orbitrap Eclipse Tribrid MS instrument (Thermo Fisher Scientific). The percentage intensity of each modification was calculated from the ratio of mass chromatogram peak areas of RNA fragments with and without the target modification using Qual Browser in Xcalibur 4.2.28.14 software (Thermo Fisher Scientific).

Nucleoside analyses of tRNAs were performed as described^{79,82}. Four micrograms of total RNA were digested into nucleosides with 0.1 U of nuclease P1 (Wako Pure Chemical Industries) in 10 mM NH₄OAc (pH 5.3) for 1 h at 37 °C, followed sequentially by 0.1 U of phosphodiesterase I (PDase I; Worthington Biochemical Corporation) in 100 mM ammonium bicarbonate, and 0.16 U of bacterial alkaline phosphatase (BAP; *E. coli* C75; Takara). For hydrophilic interaction chromatography (HILIC)/ESI-MS, digests were dissolved in 90% acetonitrile/10% water and applied to a ZIC-cHILIC column (3 µm particle size, 2.1 × 150 mm; Merck Millipore) connected to a Q Exactive Hybrid Quadrupole-Orbitrap MS instrument (Thermo Fisher Scientific) and an Ultimate 3000 LC system (Thermo Fisher Scientific). Nucleosides were separated at a flow rate of 100 µL/min in a multistep gradient comprising 90–85% solvent B from 0 to 10 min, 85–30% solvent B from 10 to 30 min with gradient curve 7, 30% solvent B for 10 min, then a return to 90% solvent B. Proton adducts of nucleosides were scanned in positive ion mode across an *m/z* range of 110–950. Xcalibur 4.4 (Thermo Fisher Scientific) was used for system operation.

Preparation of h²U-modified tRNAs by in vitro desulfuration

In vitro oxidation/desulfuration of s²U-modified tRNAs was conducted according to the literature⁵⁶ with slight modification. Human or mouse tRNA mixtures (400 µg) were dissolved in 10 mM HEPES-KOH (pH 7.4) and 100 mM H₂O₂ and incubated for 24 h at room temperature, followed by performing ethanol precipitation twice. For individual tRNAs, the isolated tRNA^{Lys} (40 µg) was dissolved in 10 mM HEPES-KOH buffer (pH 7.4) and 50 mM H₂O₂ and incubated for 24 h at room temperature, followed by ethanol precipitation twice. For complete desulfuration, the reaction was conducted twice to achieve the highest s²U-to-h²U conversion rate.

Preparation of mRNAs for cell-free translation

The pUC-T7-HCV-HA-Rluc plasmid was used for Rluc mRNA transcription. For the control, we designed pUC-T7-HCV-mHiBiT plasmid encoding a mutant HiBiT peptide with 27 amino acids (MSTNPNTLSTTGSSGWRFLRRISGGS) that does not contain any Lys, Glu, Gln, or Arg residues (encoded by AGR codons). For plasmid construction, a DNA segment spanning T7 promotor, HCV-IRES, and mutant HiBiT sequences was PCR-amplified from the pUC-T7-HCV-HA-Rluc plasmid as a template, followed by a second round of PCR to add an *Xba*I restriction enzyme site to the 3' end of the product. The amplified product was inserted into the *Hind*III and *Xba*I sites of the same vector. Finally, according to the literature^{83,84}, the beginning of the HCV-IRES sequence was mutated from MSTNPKPQRKT to MSTNPNTLSTT to generate the pUC-T7-HCV-mHiBiT plasmid by site-directed mutagenesis to avoid codons read by h²U-containing tRNAs. All primers used for plasmid construction are listed in Supplementary Data 6.

Plasmids were linearized by *Xba*I (Takara) digestion, and mRNAs were synthesized by in vitro transcription with T7 RNA polymerase using a HiScribe™ T7 High Yield RNA Synthesis Kit (NEB).

Reconstituted human cell-free translation

Reconstituted cell-free translation was carried out as described⁶¹ with modifications. Briefly, 1.26 mg/mL eEF1 complex, 100 µg/mL eEF2, 5 µg/mL eRF1/3, 0.48 µM 40S ribosomal subunit, 0.48 µM 60S

ribosomal subunit, 150 µg/mL aminoacyl-tRNA synthetase mixture, and 100 µM amino acid mixture (Promega) were mixed in a test tube in reaction buffer containing 50 mM HEPES-KOH (pH 7.5), 55.2 mM potassium acetate, 6 mM magnesium acetate, 200 µM spermidine trihydrochloride, 60 µg/mL creatine kinase, 20.6 mM creatine phosphate, 1 mM DTT, 1.36 mM ATP, and 840 µM each of GTP, UTP, and CTP. This reaction mixture was then mixed with mRNA (1.56 µM Rluc mRNA, or 1.66 µM mHiBiT mRNA) and total tRNAs with or without desulfuration (1 mg/mL) in a final volume of 10 µL, followed by incubation at 32 °C for 3 h.

Luciferase assay

One microliter of cell-free translation product of Rluc or mHiBiT mRNAs was used for measuring luciferase activity using a *Renilla* Luciferase Assay System (Promega) or a Nano-Glo HiBiT Lytic Detection System (Promega), respectively. Relative light units (RLU) of luminescence were measured by a GloMax 96 Microplate Luminometer (Promega).

Western blotting and antibodies

Cell-free translation products of Rluc mRNAs were separated by 15% SDS-PAGE and electroblotted onto an Amersham Hybond P PVDF membrane (Cytiva) using a Transblot Turbo apparatus (Bio-Rad). To detect the HA epitope of Rluc products, anti-HA-tag mAb (M180-3; MBL) was used as primary antibody at a 1:7000 dilution, and horseradish peroxidase (HRP)-conjugated donkey anti-mouse IgG (715-035-150; Jackson ImmunoResearch) was used as secondary antibody at a dilution of 1:20,000. To detect recombinant eEF-1, anti-His-tag mAb-HRP-Direct (D291-7; MBL) was used at a dilution of 1:10,000. Target proteins were detected using Pierce ECL Plus Western Blotting Substrate (Thermo Fisher Sciences) and visualized with an ImageQuant LAS4000 mini instrument (Cytiva). Band intensities were analyzed by ImageJ 1.53t.

Polysomal RNA preparation

HEK293T cells (~80% confluency) cultured in 25 dishes with 15-cm diameter were lysed with a lysis buffer comprising 20 mM Tris-HCl (pH 7.5), 150 mM KCl, 3 mM MgCl₂, 0.5% Triton X-100, 1 mM DTT, 50 µg/mL cycloheximide. The lysate was clarified by centrifugation at 13,000 × *g* for 10 min at 4 °C. The cell lysate was loaded onto a cushion buffer [20 mM Tris-HCl (pH 7.5), 150 mM KCl, 3 mM MgCl₂, 1 mM DTT, 50 µg/mL cycloheximide, and 20% sucrose] and ultracentrifuged at 35,000 rpm and 4 °C for 4 h by a Optima L-90K ultracentrifuge (Beckman) with a 70 Ti rotor (Beckman). Free RNA fraction was extracted from the supernatant using TriPure Isolation Reagent (Roche). The crude ribosome pellet was re-suspended in a ribosome binding buffer [20 mM Tris-HCl (pH 7.5), 150 mM KCl, 3 mM MgCl₂, 1 mM DTT, 50 µg/mL cycloheximide], and then loaded onto a 10–50% sucrose gradient in an SDG buffer [20 mM Tris-HCl (pH 7.5), 150 mM KCl, 3 mM MgCl₂, 1 mM DTT, 50 µg/mL cycloheximide] and ultracentrifuged at 25,000 rpm and 4 °C for 4.5 h by a Optima L-90K ultracentrifuge (Beckman) with a SW28 rotor (Beckman). The gradients were fractionated with continuous measurement of absorbance at 260 nm using a TRIAX flow cell (BioComp) and a fraction collector (Gilson, Fisher Scientific). The polysomal fraction was collected, followed by RNA precipitation by adding 3 volume of ethanol and incubating overnight at –80 °C⁸⁵. Then, polysomal RNA was extracted from the pellet using TriPure Isolation Reagent (Roche). To concentrate small RNA fraction, polysomal RNA, free RNA, and total RNA were subjected to an anion-exchange resin (DEAE Sepharose Fast Flow; GE Healthcare) with low salt buffer [10 mM HEPES-KOH (pH 7.5), 250 mM NaCl, and 2 mM DTT]. After washing the resin with the same buffer, small RNA fractions were eluted with high salt buffer [10 mM HEPES-KOH (pH 7.5), 1 M NaCl and 2 mM DTT], followed by ethanol precipitation.

Aminoacylation assay

Aminoacylation of each tRNA was performed in a 20 μ L reaction mixture containing 100 mM HEPES-KOH (pH 7.5), 50 mM KCl, 6 mM $MgCl_2$, 2 mM DTT, 2 mM ATP, 15–30 μ M ^{14}C -labeled amino acids, 20 μ M total tRNA with or without desulfuration and 750 ng aminoacyl-tRNA synthetase mixture⁶¹ at 37 °C. Before aminoacylation reaction, total tRNA was deacylated with 100 mM CHES-NaOH (pH 9.0) at 37 °C for 45 min. The concentration and specific activity of ^{14}C -labeled amino acids used in this study were as follows: 15 μ M L- $^{14}C(U)$ -Threonine (6.5 GBq/mmol; American Radiolabeled Chemicals), 15 μ M L- $^{14}C(U)$ -Arginine (11.1 GBq/mmol; American Radiolabeled Chemicals), 30 μ M L- $^{14}C(U)$ -Glutamine (9.0 GBq/mmol; PerkinElmer), 30 μ M L- $^{14}C(U)$ -Glutamic acid (8.1 GBq/mmol; American Radiolabeled Chemicals), and 15 μ M L- $^{14}C(U)$ -Lysine (12.1 GBq/mmol; PerkinElmer). At each time point, the reaction was quenched by spotting 4 μ L aliquots of reaction mixture onto a Whatman 3MM Chr filter paper (Cytiva). The filter paper was pre-soaked in a 5% trichloroacetic acid solution before use. The spotted filter paper was washed three times for 15 min each in ice-cold 5% trichloroacetic acid, washed with ethanol and dried under a heat lamp. Acid-insoluble radioactivity of aminoacyl-tRNAs was measured by a liquid scintillation analyzer (PerkinElmer).

Ribosomal A-site tRNA binding

tRNA binding to the A-site on the ribosome was carried out as described^{86,87} with modifications. mRNAs containing the Shine-Dalgarno sequence and an AAR codon (R=A or G) codons or a UUC codon (negative control) at the A-site (Supplementary Data 6) were chemically synthesized. Human cytoplasmic tRNA^{Lys} bearing mcm⁵s²U or mcm⁵h²U was dephosphorylated in buffer containing 10 mM HEPES-KOH (pH 7.5) and 0.008 units of bacterial alkaline phosphatase (*E. coli* C75; Takara) at 55 °C for 30 min, followed by gel purification of the intact tRNAs. The dephosphorylated tRNAs were 5' phosphorylated with [γ -³²P]ATP (PerkinElmer) by T4 polynucleotide kinase (NEB), followed by gel purification of the 5'-³²P-labeled tRNAs. Radioactivity of labeled tRNAs was measured by Cherenkov counting using a liquid scintillation analyzer (PerkinElmer). The specific activities of labeled tRNAs were adjusted to 20,000 cpm/pmol by adding respective cold tRNA^{Lys} with or without desulfuration. First, *E. coli* 70S ribosomes (1.6 pmol) were incubated with *E. coli* tRNA^{fMet} (25 pmol) and mRNA (20 pmol) at 37 °C for 40 min in binding buffer containing 50 mM HEPES-KOH (pH 7.5), 120 mM KCl, 6.5 mM $Mg(OAc)_2$, 1 mM DTT, and 0.5 mM spermine to occupy the P-site of the ribosome. Next, 0.1, 0.25, or 0.5 pmol of the ³²P-labeled tRNA^{Lys} was added to the mixture and incubated at 37 °C for 15 min, then at 0 °C for 30 min. The mixture was spotted onto double-layered Amersham Protran nitrocellulose membrane (Cytiva) and Hybond-N⁺ nylon membrane (Cytiva) using a 96-well dot-blot vacuum apparatus (SAMPLATEC, Japan). The membranes were washed twice with 200 μ L of binding buffer, dried, exposed to an imaging plate, and visualized on an FLA-7000 image analyzer (Fujifilm). The amount of bound tRNA was quantified using Multi Gauge V3.0 (Fujifilm).

Single-particle cryo-EM analysis of ribosomes with human tRNAs

The cryo-EM single-particle analysis of the ribosome–tRNA complexes was basically performed in accordance with our previous studies^{5,78,86,88}. *E. coli* 70S ribosomes were prepared from the MRE600 strain as previously described⁵, resuspended in RBS buffer containing 20 mM HEPES-KOH (pH 7.6), 10 mM $Mg(OAc)_2$, 30 mM NH_4Cl , and 6 mM β -mercaptoethanol. Methionylation and formylation of tRNA^{fMet} were carried out as described previously⁸⁹. To form a ribosome complex, *E. coli* 70S ribosomes were mixed with mRNA and P-site tRNA in a mixture containing 20 mM HEPES-KOH (pH 7.6), 10 mM $Mg(OAc)_2$, 30 mM NH_4Cl , 6 mM β -mercaptoethanol, 100 nM 70S ribosomes, 500 nM mRNA, and 500 nM *E. coli* fMet-tRNA^{fMet}, followed by incubation at 37 °C for 15 min. Next, 500 nM of human

tRNA^{Lys} was added as A-site tRNA and the mixture was incubated at 37 °C for 30 min, followed by further incubation on ice for 1 h. Grid preparation was carried out as previously described⁵.

Ribosome particle images were collected using a Titan Krios transmission electron microscope (Thermo Fisher Scientific) operated at an accelerating voltage of 300 kV and equipped with a K3 camera (Gatan). Automated image acquisition was performed at a nominal magnification of $\times 105,000$ with a defocus range from 0.5 to 2.5 μ m using a K3 direct electron detector (Gatan) in CDS-counting mode (0.8285 $\text{\AA}/\text{pixel}$; Supplementary Data 3). The total exposure of electrons at the specimen level was $\sim 50 \text{ e}^-/\text{\AA}^2$, and dose was fractionated into 48 frames (Supplementary Data 3).

Images were recoded as movie micrographs and image processing was performed using RELION-3.1.3⁹⁰. Details of image processing for each ribosome complex are described in Supplementary Fig. 5. Movies were motion-corrected with the program implemented in RELION, and contrast transfer function (CTF) parameters were estimated with CTFFIND-4.1⁹¹. Ribosome particles were automatically picked by crYOLO-1.8.2⁹². The obtained particles were initially subjected to 2D classification to discard ice contamination, and the remaining particles were then auto-refined to reconstruct a consensus map. Subsequently, based on the alignment of consensus reconstruction, 3D classification was performed. The subgroup of 70S ribosome particles was auto-refined and further classified using a mask around A-site tRNA to discard ribosomes with a vacant A-site. Subgroups of A-site tRNA-bound ribosomes were further auto-refined followed by CTF refinement, Bayesian polishing and another round of 3D refinement and postprocessing. The final resolution reached 2.26–3.18 $\text{\AA}$. The angular distribution of particle projections was calculated by RELION-3.1.3⁹⁰. The local resolution of each map was estimated by RELION-3.1.3⁹⁰ (Supplementary Fig. 6a–d).

The high-resolution structure of the *E. coli* 70S ribosome complexed with P-site tRNA^{fMet} and A-site tRNA (PDB: 7y7e)⁵ was used as a starting model and rigid body-fitted to the obtained cryo-EM maps. Models of mRNA and A-site tRNA were manually constructed using COOT-0.9.5⁹³, and tRNA modifications were manually inserted into the model. The chemical structure of mcm⁵h²U was modeled using eLBOW⁹⁴. These initial models were real space-refined to cryo-EM maps by the phenix.real space refine routine⁹⁵. These models were corrected using UCSF Chimera-1.14⁹⁶ and COOT-0.9.5⁹³, followed by a final phenix.real space refine routine using Phenix-1.18⁹⁵. Refined final models were validated using MOLPROBITY⁹⁷. Map vs. model FSC values were calculated using Phenix-1.18⁹⁵ (Supplementary Fig. 6e, f). Graphical figures were prepared using UCSF Chimera-1.14⁹⁶ or UCSF ChimeraX 1.3⁹⁸.

Synonymous codon usage bias analysis

Gene lists associated with the Gene Ontology (GO) term “response to oxidative stress” (GO:0006979) were retrieved using AmiGO 2 (<http://amigo.geneontology.org>). Gene names were mapped to corresponding NCBI RefSeq accessions using the UniProt ID mapping tool (<https://www.uniprot.org/id-mapping>). The coding sequences (CDS) corresponding to the RefSeq identifiers were then downloaded from NCBI in FASTA format. For each sequence, valid open reading frames (ORFs)—defined as regions starting with a canonical ATG initiation codon and terminating at the first in-frame stop codon (TAA, TAG, or TGA)—were extracted from the FASTA files.

Relative synonymous codon usage (RSCU) values were calculated using a custom Python script with Biopython v1.81, based on the observed versus expected frequencies of each synonymous codon.

Figure preparation

All figures were prepared using Canvas X 20 (Nihon Poladigital K.K.). Chemical structures were drawn with ChemDraw 23.0.1 (PerkinElmer). Graphs were generated by GraphPad Prism 7.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

Publicly available datasets from Protein Data Bank (7y7e) were used for atomic model building and comparison. Cryo-EM maps and atomic coordinates of the reported structures were deposited in the Electron Microscopy Data Bank (EMDB) and Protein Data Bank, respectively, with the following accession codes: EMD-61706 and 9JPM (tRNA^{Lys} with mcm⁵s²U34 and the AAA codon); EMD-61708 and 9JPO (tRNA^{Lys} with mcm⁵h²U34 and the AAA codon); EMD-61709 and 9JPP (tRNA^{Lys} with mcm⁵s²U34 and the AAG codon); EMD-61720 and 9JQ2 (tRNA^{Lys} with mcm⁵h²U34 and the AAG codon). Source data are provided with this paper.

Code availability

The code generated in this study has been deposited in Zenodo under <https://doi.org/10.5281/zenodo.18289446>.

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Author contributions

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Competing interests

The authors declare no competing interests.

Additional information

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